Business Question and Visualization Report

Date	24 September 2026
Team ID	
Project Name	EV Market Analytics Dashboard
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

Business Questions, Code and Visualisations

Database: EV_DB | Warehouse: EV_WH | Dashboard tool: Streamlit (connected to Snowflake)

1. Which vehicle categories contribute the most to annual sales?

Code:

```
SELECT VEHICLE_CATEGORY,
       SUM(TOTAL_SALES) AS TOTAL_SALES
FROM EV_DB.MART.VW_CATEGORY_SALES
GROUP BY VEHICLE_CATEGORY
ORDER BY TOTAL_SALES DESC;
```

Visualisation: Bar chart: total sales by vehicle category

[Screenshot of visualisation]

2. How have total EV sales changed year over year?

Code:

```
SELECT SALES_YEAR,
       SUM(ANNUAL_SALES) AS TOTAL_SALES,
       ROUND(100 * (SUM(ANNUAL_SALES) - LAG(SUM(ANNUAL_SALES)) OVER (ORDER BY SALES_YEAR))
             / NULLIF(LAG(SUM(ANNUAL_SALES)) OVER (ORDER BY SALES_YEAR), 0), 2) AS YOY_GROWTH_PCT
FROM EV_DB.CURATED.EV_DATA_CLEAN
GROUP BY SALES_YEAR
ORDER BY SALES_YEAR;
```

Visualisation: Line chart: sales trend with year-over-year growth %

[Screenshot of visualisation]

3. Which manufacturers hold the largest market share?

Code:

```
SELECT MANUFACTURER, TOTAL_SALES, MARKET_SHARE_PCT
FROM EV_DB.MART.VW_MANUFACTURER_SHARE
ORDER BY TOTAL_SALES DESC
LIMIT 10;
```

Visualisation: Bar / donut chart: top 10 manufacturers by market share

[Screenshot of visualisation]

4. How does customer satisfaction vary across vehicle categories?

Code:

```
SELECT VEHICLE_CATEGORY,
       ROUND(AVG(CUSTOMER_RATING), 2) AS AVG_RATING,
       COUNT(*)                       AS MODEL_COUNT
FROM EV_DB.CURATED.EV_DATA_CLEAN
WHERE CUSTOMER_RATING IS NOT NULL
GROUP BY VEHICLE_CATEGORY
ORDER BY AVG_RATING DESC;
```

Visualisation: Bar chart: average customer rating by category

[Screenshot of visualisation]

5. Which battery technologies are most widely adopted?

Code:

```
SELECT BATTERY_TYPE,
       SUM(TOTAL_SALES) AS TOTAL_SALES,
       ROUND(100 * SUM(TOTAL_SALES) / SUM(SUM(TOTAL_SALES)) OVER (), 2) AS SHARE_PCT
FROM EV_DB.MART.VW_BATTERY_SUMMARY
GROUP BY BATTERY_TYPE
ORDER BY TOTAL_SALES DESC;
```

Visualisation: Donut chart: sales share by battery type

[Screenshot of visualisation]

6. How does driving range compare across battery types and vehicle categories?

Code:

```
SELECT VEHICLE_CATEGORY,
       BATTERY_TYPE,
       ROUND(AVG(DRIVING_RANGE_KM), 1) AS AVG_RANGE_KM
FROM EV_DB.CURATED.EV_DATA_CLEAN
WHERE DRIVING_RANGE_KM IS NOT NULL
GROUP BY VEHICLE_CATEGORY, BATTERY_TYPE
ORDER BY VEHICLE_CATEGORY, AVG_RANGE_KM DESC;
```

Visualisation: Grouped bar chart / heatmap: average range by category and battery type

[Screenshot of visualisation]

7. Which countries lead EV manufacturing?

Code:

```
SELECT MANUFACTURING_COUNTRY,
       COUNT(DISTINCT MANUFACTURER) AS MANUFACTURER_COUNT,
       SUM(ANNUAL_SALES)            AS TOTAL_SALES
FROM EV_DB.CURATED.EV_DATA_CLEAN
WHERE MANUFACTURING_COUNTRY <> 'Unknown'
GROUP BY MANUFACTURING_COUNTRY
ORDER BY TOTAL_SALES DESC;
```

Visualisation: Map / bar chart: manufacturers and sales by country

[Screenshot of visualisation]

8. Is a larger battery capacity linked to a longer driving range?

Code:

```
SELECT BATTERY_CAPACITY_KWH,
       DRIVING_RANGE_KM,
```

```
       BATTERY_TYPE,
       VEHICLE_CATEGORY
FROM EV_DB.CURATED.EV_DATA_CLEAN
WHERE BATTERY_CAPACITY_KWH IS NOT NULL AND DRIVING_RANGE_KM IS NOT NULL;

-- strength of the relationship
SELECT ROUND(CORR(BATTERY_CAPACITY_KWH, DRIVING_RANGE_KM), 3) AS CORRELATION
FROM EV_DB.CURATED.EV_DATA_CLEAN;
```

Visualisation: Scatter plot: battery capacity vs driving range, coloured by battery type

[Screenshot of visualisation]